

Alex Shao

alex.w.shao@gmail.com · [LinkedIn](#) · [GitHub](#) · [Project Portfolio](#)

Education

University of California, Berkeley

Pure Mathematics

August – December 2024, August 2026 – Present

GPA: 3.73 | Major GPA: 3.9

Foothill College

Mathematics and Computer Science

September 2023 – June 2024

Dean's List, High Honors, GPA: 3.95

Artificial Intelligence Professional Program, Stanford University

February 2024, November 2025

Courses: Machine Learning with Graphs (XCS224W) [\[cert\]](#), Reinforcement Learning (XCS234) [\[cert\]](#)

Skills

Languages: Python, C, C++, CUDA, Rust

Embedded & Systems: STM32, ROS2, EtherCAT, real-time Linux, Arduino

Robotics & Simulation: Isaac Sim/Lab, MuJoCo, real-time robot control, MATLAB, SolidWorks

Machine Learning: PyTorch, reinforcement learning, imitation learning, skill discovery (SAC), HPC

Work & Research Experience

Symbiokinetics Inc., Sunnyvale, CA

Researcher & Robotician

January 2026 – Present

- Real-time control for safe physical human-robot interaction.
- Robot learning for contact-rich manipulation and skill transfer.

Embedded Software Engineer

January 2025 – December 2025

- Designed and integrated a hardware-software demo from concept to working build in 2 months: enclosure, electronics, peripherals, and system integration.
- Wrote STM32 baremetal firmware for data acquisition, device communication, and embedded networking.
- Built inter-process and cross-machine communication, custom ROS2 message types and UDP transport over Ethernet, to offload compute from the primary real-time controller.
- Implemented real-time EtherCAT motor-drive control and device state machine.

Intern

June 2023 – June 2024

- Simulated backdrivability and haptic interaction in MATLAB/CoppeliaSim.
- Designed and 3D-printed custom carbon-fiber end effectors and sensor mounting fixtures for Dexter HDI and Amber B1.

Bongard AI, Palo Alto, CA — Research Intern

June 2023 – January 2024

- Developed a massively parallelized batched K-Means clustering algorithm in Numba and CUDA C.
- CUDA C implementation achieved a **1600× speedup** compared to scikit-learn KMeans.

Biomimetic Millisystems Lab, UC Berkeley — Researcher

September – December 2024

- Developed a repeatable heat-seal process for bonding LDPE pneumatic pouches to pressure valves, eliminating chronic seal failures under repeated strain for soft robot manufacturing.

Projects

Generalized Compliant Control via Skill Discovery

2026

- Implemented Structured Unsupervised Skill Discovery (SUSD) combined with compliant control to produce auto-tuned, generalized motor behavior in simulation.

GPU-Accelerated Batched K-Means [\[Colab\]](#)

2023

- Published CUDA C / Python implementation on [Towards Data Science](#), accumulating **10k+ reads**.